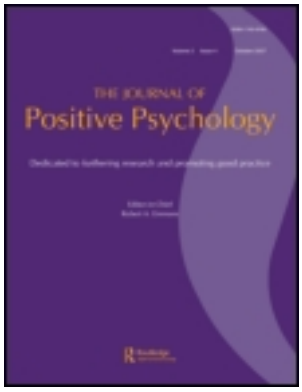
This article was downloaded by: [Sarah Schnitker]
On: 26 June 2012, At: 19:09
Publisher: Routledge
Informa Ltd Registered in England and Wales Registered Number: 1072954 Registered office: Mortimer House, 37-41 Mortimer Street, London W1T 3JH, UK



The Journal of Positive Psychology: Dedicated to furthering research and promoting good practice
Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/rpos20

An examination of patience and well-being

Sarah A. Schnitker [a]
[a] School of Psychology, Fuller Theological Seminary, Pasadena, CA 91106, USA

Available online: 26 Jun 2012

To cite this article: Sarah A. Schnitker (2012): An examination of patience and well-being, The Journal of Positive Psychology: Dedicated to furthering research and promoting good practice, 7:4, 263-280

To link to this article: http://dx.doi.org/10.1080/17439760.2012.697185

PLEASE SCROLL DOWN FOR ARTICLE

Full terms and conditions of use: http://www.tandfonline.com/page/terms-and-conditions

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden.

The publisher does not give any warranty express or implied or make any representation that the contents will be complete or accurate or up to date. The accuracy of any instructions, formulae, and drug doses should be independently verified with primary sources. The publisher shall not be liable for any loss, actions, claims, proceedings, demand, or costs or damages whatsoever or howsoever caused arising directly or indirectly in connection with or arising out of the use of this material.

The Journal of Positive Psychology
Vol. 7, No. 4, July 2012, 263–280

Routledge
Taylor & Francis Group

An examination of patience and well-being

Sarah A. Schnitker*

School of Psychology, Fuller Theological Seminary, Pasadena, CA 91106, USA

(*Received 18 May 2012; final version received 22 May 2012*)

Patience is the propensity to wait calmly in the face of frustration or adversity. The new 3-Factor Patience Scale (Study 1) measures three types of patience (*interpersonal*, *life hardship*, and *daily hassles patience*), which differentially relate to well-being and personality. In Study 2, goal pursuit and achievement mediated the relation between patience and well-being. Participants rated 10 personal goals on 15 dimensions of goal pursuit (e.g. patience enacted, difficulty, achievement satisfaction, effort). Patience facilitated goal pursuit and satisfaction especially in the face of obstacles. In Study 3, participants took part in a training program designed to increase trait patience. The program led to increased patience, decreased depression, and increased positive affect relative to a control condition, suggesting that patience may be modifiable.

Keywords: patience; well-being; goals; virtue

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

Introduction

Patience has long been upheld as a character strength and desirable personality trait that promotes human flourishing and well-being. Familiar maxims such as 'good things come to those who wait' exemplify the desirability of the trait, and moral philosophers and religious writers have emphasized the importance of developing patience to achieve the 'good life' (Harned, 1997). Although touted as an important trait for millennia, patience is only recently receiving attention as a topic of empirical investigation in psychology.

What is patience?

In the most basic sense, patience is the propensity of a person to wait calmly in the face of frustration, adversity, or suffering. Patience is enacted across a wide range of circumstances and timeframes. It is enacted (or not!) in mundane activities, such as waiting in traffic, as well as in more significant and long-term situations, such as parenting or dealing with a serious illness. Although it often involves a temporal or waiting component, patience is also called forth in situations with no direct focus on time (e.g. dealing with a difficult person).

Typically construed as a disposition (e.g. 'She is a patient person'), patience can also be viewed as a state (e.g. 'She waited patiently'). These two conceptualizations of patience are intertwined (people with high dispositional patience will experience more discrete states of patience), so it is important to consider both of them as meaningful features. Demonstration of genuine patience depends on both behavioral (i.e. waiting) and emotional (i.e. low arousal positive affect and notable absence of high arousal negative affect) components.

Why study patience?

Situations that require patience are fundamental to human experience. People regularly encounter frustrating circumstances where they may or may not exhibit patience (e.g. parenting, working with a difficult boss). Waiting is an unavoidable part of life; a poll reported in *The Daily Telegraph* found that the average Briton spends 5 h and 35 min each month – or six months of an average lifetime – standing in line (Britons Spent Six Months Queuing, 2009). Research shows that daily hassles and frustrations have a negative impact on physical health and well-being, perhaps of a greater magnitude than major life events (DeLongis, Coyne, Dakof, Folkman, & Lazarus, 1982). Given the pervasiveness of waiting and frustrating stimuli in daily life, research examining individual differences in how people react to such circumstances is needed to maximize human functioning. Initial evidence shows that patience is positively correlated with subjective well-being, positive coping, virtues, and thriving (Schnitker & Emmons, 2007). Research on personality and social processes underlying the relation between patience and well-being has

*Email: sschnitker@fuller.edu

ISSN 1743–9760 print/ISSN 1743–9779 online
© 2012 Taylor & Francis
http://dx.doi.org/10.1080/17439760.2012.697185
http://www.tandfonline.com

applications to improving quality of life in a plethora of settings.

Patience and well-being

Patience is hypothesized to affect both hedonic and eudaimonic well-being – two long-established notions representing distinct, but overlapping, paradigms to understand 'the good life'. Hedonism reflects the idea that well-being consists of maximizing pleasure or happiness and minimizing negative emotions (Ryan & Deci, 2001). Patience directly influences hedonic well-being as it buffers against emotions in stressful situations, allows the person to cope more adaptively with frustrations, and facilitates positive interpersonal interactions. Though often misconstrued to concern only physical pleasures, hedonism also comprises the perspective that happiness arises from the achievement of goals (Diener, Sapyta, & Suh, 1998). Thus, patience may indirectly affect hedonic well-being by facilitating goal attainment, which increases positive emotions and life satisfaction.

A eudaimonic view of well-being maintains that the 'good life' lies in the realization of human potential and the development of virtue (e.g. Ryff & Singer, 1998). Self-determination theory defines eudaimonic well-being in terms of intrinsic motivation, integrity, well-being, vitality, and self-congruence, which are realized by the fulfillment of autonomy, competence, and relatedness needs (Ryan & Deci, 2000). Patience is proposed to increase eudaimonic well-being by facilitating the realization of goals that promote the fulfillment of these basic needs.

Prior research related to trait patience

Patience was framed by Curry, Price, and Price (2008) as low discounting rates across time. They operationally define the patient person as one who is less likely to discount the benefits of a reward if forced to wait for it, and they found that patience positively correlated with increased cooperation in a public-good game. This conceptualization of patience highlights an important distinction between patience and delay of gratification. Although related to patience, delay of gratification is distinguished as involving an overt choice between a smaller reward now versus a larger reward later. In contrast, patience is often demonstrated in situations where there is no choice about whether or not to wait – the only choice is *how* one waits.

Impatience has been examined in research on the *Impatience/Irritability* factor of the Type A personality. Spence, Helmreich, and Pred (1987) found that those scoring high on the *Impatience/Irritability* factor reported a greater number of physical complaints, but studies indicate that it is hostility and hard-driving competitiveness, rather than impatience alone, that drives the association between Type A personality and cardiovascular disease (Miller, Smith, Turner, Guijarro, & Hallet, 1996). Although findings on impatience inform an understanding of patience, it is important to note that impatience and patience are not mirror opposites. Patience is better conceptualized as a mean between two poles of behavior. According to Aristotle (Rorty, 1980), a virtuous action is always an intermediate state between two extremes of vice that are reflected in excess and deficiency. In this case, patience is a mean between the excess of *recklessness* and the deficiency of *sloth*.

Prior research related to state patience

Researchers have uncovered several key aspects of situations that affect momentary patience and impatience. First, marketing researchers have examined impatience in customer service ratings. An impatient state will arise and service satisfaction will decrease if a person perceives that the time spent waiting has surpassed expectations for waiting (Groth & Gilliland, 2006). Moreover, when a wait is attributed to the service provider, customers exhibit even greater impatience and dissatisfaction (Rose, Meuter, & Curran, 2005).

People differ in their 'wait-time construals', or how long they believe they should have to wait in a specific situation. Several factors (e.g. situation factors, past experiences, basic personality disposition, socio-cultural norms, etc.) lead to the formation of these construals. Roy, Cristenfeld, and McKenzie (2005) found that people especially underestimate how long it will take them to complete a task when they are highly motivated to finish quickly. In studies using a word game (Buehler, Griffin, & MacDonald, 1997) or origami task (Byram, 1997), participants who offered a monetary incentive to perform faster underestimated how long it would take them to complete the task compared to participants who were not offered an incentive; however, the incentive had no effect on how quickly the participants actually completed the task. Thus, when a person wants an activity to occur quickly, construals of how long the situation should last shorten, and patience will most likely decrease.

Time estimation may be involved in state patience. Situations that emphasize temporally salient information will increase time estimates, which may decrease patience. Situations that lead to absorption and engagement of attention (i.e. increased mindfulness) will decrease time estimates, thus increasing patience (Glicksohn, 2001).

Even after people's situation construals are violated and they recognize that they are waiting or enduring a frustration longer than expected, they may still

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

respond patiently by utilizing emotion regulation techniques. For example, patient people may utilize reappraisals (Clore & Robinson, 2000) to change their expectations and feel more patient. We hypothesize that if people are taught such emotion regulation strategies, they will be able to increase their ability and propensity to wait patiently.

Preliminary understanding of patience: Correlates of patience

Considering patience as a trait and character strength, Schnitker and Emmons (2007) created The Patience Scale-10 (PS-10) to measure a person's self-evaluation of patience (e.g. 'Most people would say I am a patient person', or 'Waiting in lines does not bother me') and attitudes about the importance of patience (e.g. 'I believe that when it comes to getting along with others, patience is an important factor'). In response to claims that patience is merely an amalgam of other virtues (Peterson & Seligman, 2004), Schnitker and Emmons (2007) established the discriminant validity of patience. In a regression of patience on the 24 character strengths of the Values in Action Inventory of Strengths (VIA-IS; Peterson, Park, & Seligman, 2005), the 24 strengths combined accounted for only 26% of the variance in patience scores, indicating that patience is not reducible to other virtues. In a regression on the three virtues specifically hypothesized by Peterson and Seligman (2004) to compose patience (i.e. persistence, open-mindedness, and self-regulation), only open-mindedness significantly predicted patience, and the three strengths combined accounted for only 9% of the variance in patience scores.

Schnitker and Emmons (2007) distinguished patience from basic personality traits while also providing evidence of convergent validity. In a regression of patience on the Big Five (John, Donahue, & Kentle, 1991), the five factors accounted for 25% of the variance in patience scores. Patience was correlated with higher Agreeableness, higher Openness, and lower Neuroticism. They also found a robust relation between patience and well-being. Higher patience correlated with decreased negative affect, decreased depression, and lower incidence of health problems. Patience positively correlated with prosocial traits (e.g. empathy) and character strengths of the VIA-IS (Peterson et al., 2005), namely strengths of Temperance (Forgiveness $r = 0.38$, Prudence $r = 0.29$) and Justice (Equity $r = 0.41$, Leadership $r = 0.33$).

However, the PS-10 suffered several weaknesses as a measure of patience. First, the confirmatory factor analysis of the PS-10 reported by Schnitker and Emmons (2007) did not show good fit with a root mean square error of approximation (RMSEA) of 0.098 (an RMSEA less than 0.10 is considered passable, but good fit would be indicated with an RMSEA at or less than 0.05; see Byrne, 1998). Second, the PS-10 is limited to obtaining a person's beliefs about the importance of patience and a general self-evaluation of patience. The scale does not capture theoretical distinctions and nuances of patience that are important for well-being outcomes. For example, patience in dealing with chronic illness (life hardship patience) differs considerably from patience when dealing with a recalcitrant child (interpersonal patience) or when stuck in traffic (daily hassles patience).

Research objectives

Given that the heretofore described research summarizes nearly all published work on patience, many opportunities for data collection and theory building abound. To limit our scope of inquiry, we focus on three main objectives in our research.

Objective 1 – Establish construct validity and distinguish different types of patience. As expounded by John and Soto (2007), we seek to amass evidence of construct validity for patience and build a nomological network around the trait. In this vein, we present a more valid and reliable new patience scale, capturing distinctive dimensions of patience based on the nature of the patience-eliciting stimuli. We hypothesize that the antecedents and well-being outcomes from these three types of patience will differ. We develop a new scale measuring these three factors in Study 1, and we construct a multitrait–multimethod matrix (Campbell & Fiske, 1959) with the collection of informant reports in Study 2.

Objective 2 – Test the directionality and causality of the patience-well-being relation. The second objective of the present research is to better understand the relation between patience and well-being. Previous findings on the relation between patience and well-being (Schnitker & Emmons, 2007) were ascertained from concurrent correlational data and do not provide insight into the directionality or causality of effects. Collection of longitudinal data in Study 2 and experimental data in Study 3 correct this deficit.

Addressing possible negative outcomes. Although patience is theorized to engender beneficial effects, it is possible that the trait may lead to negative outcomes. It has been suggested that patience leads to passivity and that too much patience can be injurious (Burke, 1769). Block (1996) found negative outcomes for people at the extreme high end of the ability to delay gratification, and Robins, John, Caspi, Moffitt, and Stouthamer-Loeber (1996) found that boys with an overcontrolled personality type (compared to those with resilient and undercontrolled types) were prone to internalizing problems. Like these positive traits, can too much patience also have negative effects?

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

Variables that may point to the deleterious effects of patience, such as assertiveness and passivity in goal pursuit, are included in the succeeding studies to specifically test for possible negative effects.

Objective 3 – Examine mediation of the relation between patience and well-being by goal pursuit and achievement. We endeavor to understand the personality processes mediating the relation between patience and well-being, particularly looking at the processes of goal pursuit and achievement. We hypothesize that patience will increase goal achievement, which will increase well-being.

The goal striving literature has traditionally concerned itself with the relation between goals and well-being outcomes. Palys and Little (1983) maintain that goal enjoyability, difficulty, social support, and temporal importance are associated with life satisfaction. Sheldon, Ryan, Deci, and Kasser (2004) provide evidence that goal content and motives affect quality of life in a 1-year prospective study of change in well-being. Within the goal framework, one way to conceive of patience is as a coping tool in response to goal delay or potential failure. Echteld, Elderen, and Kamp (2001) highlight the adaptive function of avoidant coping when a person has no sense of control over a situation. Approach coping in an uncontrollable situation may cause high goal disturbance and high perceptions of stress caused by that disturbance – a combination that leads to the lowest quality of life scores. Although patience is not tantamount to avoidant coping, this research illuminates possible pathways from patience to beneficial outcomes.

Brandtstadter and Renner's (1990) findings on tenacious goal pursuit and flexible goal adjustment are also informative. Tenacious goal pursuit refers to the assimilative process of transforming external circumstances to fit personal preferences, and flexible goal adjustment refers to the accommodative tendency to adjust personal preferences and goal orientations to fit situational constraints (akin to patience). Brandtstadter and Renner report that *both* tenacious goal pursuit and flexible goal adjustment predict high life satisfaction and low depression, and *both* are positively correlated with internalized control beliefs. Brandtstadter and Rothermund (2002) findings support the importance of utilizing accommodative strategies in uncontrollable circumstances: the negative relation between measures of control and depression disappear in circumstances that are impervious to change (Wolk, 1976).

Study 1: Creating a three factor patience scale

The purpose of this study is to create and validate a measure that distinguishes between *interpersonal patience*, *life hardship patience*, and *daily hassles patience*.

Method

Participants

Participants were 389 UC Davis undergraduates (296 females, 85 males; mean age = 19.8 years) recruited through the psychology subject pool who received course credit. Participants were primarily Asian American (43%) and Caucasian (24%), though other ethnic minority groups were represented (11% Hispanic, 3% African American, 2% Middle Eastern, 15% other).

Measures

Patience items. Participants were administered 40 items hypothesized to measure interpersonal patience (e.g. 'When someone is having difficulty learning something new, I will be able to help them without getting frustrated or annoyed'), life hardship patience (e.g. 'I am able to wait-out tough times'), and daily hassles patience (e.g. 'Although they're annoying, I don't get too upset when stuck in a traffic jam'). Items were rated on a 7-point Likert scale (1 = *very much unlike me*; 7 = *very much like me*).

PS-10. The Schnitker and Emmons (2007) PS-10 was used to assess beliefs about the importance of patience and self-evaluation of patience. The 10 items were rated on a 7-point Likert scale (1 = *very much unlike me*; 7 = *very much like me*).

Well-being indicators

CES-D. The Center for Epidemiological Studies Depression Scale (CES-D; Radloff, 1977) was used to measure depression with 20 self-report items. Participants rated (1 = *rarely or none of the time*; 4 = *most or all of the time*) how often they exhibited depressive symptoms during the week prior to completing the questionnaire.

Rosenberg Self-Esteem Inventory. Ten items of the Rosenberg Self-Esteem Inventory (Rosenberg, 1965) were rated on a 1 = *strongly disagree* to 4 = *strongly agree* scale.

Satisfaction with Life Scale. This 5-item scale (Diener, Emmons, Larsen, & Griffin, 1985) was used to measure global life satisfaction (1 = *strongly disagree*; 7 = *strongly agree*).

Hope. Snyder et al.'s (1991) 8-item Hope Scale was used to measure the dispositional trait of hope (1 = *definitely false*; 8 = *definitely true*).

***R-UCLA Loneliness Scale* (*Short Form*).** This 3-item scale of general loneliness (Hughes, Waite, Hawkley, & Cacioppo, 2004) asks subjects to rate how often they feel isolated, left out, and lacking companionship (1 = *hardly ever*, 2 = *some of the time*, and 3 = *often*).

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

Personality measures

Big Five Inventory (BFI). The 44-item BFI (John et al., 1991) measured the personality factors of Extraversion, Agreeableness, Conscientiousness, Neuroticism, and Openness to Experience on a 5-point Likert scale (1 = *disagree strongly*; 5 = *agree strongly*).

Assertiveness. The Rathus Assertiveness Schedule (1973) was used to measure assertive behaviors. The 30 items of this measure are scored on a −3 to +3 scale (−3 = *very unlike me*; +3 = *very like me*). Example items include 'Anyone attempting to push ahead of me in line is in for a good battle' and 'I have a hard time saying no' (reverse scored).

Mindful Attention Awareness Scale (MAAS). Brown and Ryan's (2003) 15-item scale was used to measure present and mindful engagement (1 = *almost always*; 6 = *almost never*).

Attachment. The Experiences in Close Relationships Scale (ECR; Brennan, Clark, & Shaver, 1998) is a 36-item measure of adult attachment (1 = *disagree strongly*; 7 = *agree strongly*) indicating secure, insecure-anxious, or insecure-avoidant attachment in relationships.

Procedure

Participants took the survey online using the Concentus Assessment survey platform. They completed the preliminary patience items and then completed the other personality and well-being measures. Two weeks later, participants were asked via e-mail to take a second online survey. Participants ($N = 317$) again completed the preliminary patience items, as well as other measures from the Time 1 survey, to establish test–retest reliability.

Results

Initial analyses of scale construction: Exploratory factor analyses

Exploratory factor analyses (EFAs, maximum-likelihood extraction/promax rotation) were run on the initial pool of patience items. Examination of the scree plot revealed one large factor and two or three additional factors before the plot leveled off (Figure 1a). Factor loadings for 1-, 2-, 3-, and 4-factor solutions were examined; the 3-factor solution demonstrated the most simple structure and greatest interpretability. Factors appeared to represent interpersonal, life hardship, and daily hassles patience, respectively.

The pool of 40 items was reduced to 11. Items loading less than 0.30 on a factor, with content loading equally on multiple factors, or with content that did not match the theoretical conceptualization of the factors were eliminated. Reliability and item analyses were employed to remove additional items that did not correlate with other items or were redundant. An EFA of these 11 items produced three factors (*interpersonal*, *life hardship*, and *daily hassles patience*) with eigenvalues greater than one (4.23, 1.25, 1.08), accounting for 38.48%, 11.35%, and 9.85% of the variance, respectively. However, the scree plot supported a 1-factor solution (Figure 1b). Thus, the goodness-of-fit for the 1-, 2-, and 3-factor models were tested.

Confirmatory factor analyses of EFA factor structure

Confirmatory factor analyses were run with 1-, 2-, and 3-factor models (both orthogonal and non-orthogonal). Models were tested using maximum-likelihood estimation in LISREL Version 8.54. Fit statistics for all the models are shown in Table 1. As expected, the non-orthogonal 3-factor model best fit the data, $X^2(41, N = 359) = 107.30$, $p < 0.001$. Akaike's information criterion (AIC) was 158.25, and the comparative fit index (CFI) was 0.96. These indices and an RMSEA of 0.054 indicate fair to good fit (Byrne, 1998). The next best-fitting model, the 3-factor orthogonal model, had poorer indicators of fit (AIC = 226.64, RMSEA = 0.094, CFI = 0.90), so we adopted the 3-factor correlated model with confidence. Figure 2 depicts the adopted model and parameter estimates.

Reliability and scale descriptives

The appendix shows the final 3-Factor Patience Scale (3-FPS). As can be seen in Table 2, the scale demonstrates adequate internal reliability ($\alpha = 0.66$–0.80) and an average 2-week test–retest reliability of 0.66. Scores are normally distributed and do not systematically differ by gender or ethnicity. Scores on the 3-FPS correlate moderately to highly with the self-evaluation factor of the PS-10 ($r = 0.66, 0.38, 0.66$ for *interpersonal*, *life hardship*, and *daily hassles patience*, respectively) and moderately with beliefs about the importance of patience ($r = 0.43, 0.20, 0.33$, respectively).

Patience and personality: External validity and possible antecedents of patience

Big Five. The three patience factors have significant zero-order correlations with four of the five Big Five traits (Agreeableness, Conscientiousness, Neuroticism, and Openness; $p < 0.01$). Of note, *interpersonal patience* has a strong correlation with Agreeableness ($r = 0.60$), whereas *life hardship* and *daily hassles patience* have moderate correlations ($r = 0.22, 0.30$). Table 3 shows each of the factors of the 3-FPS regressed on the Big Five. When *interpersonal patience* is regressed on the

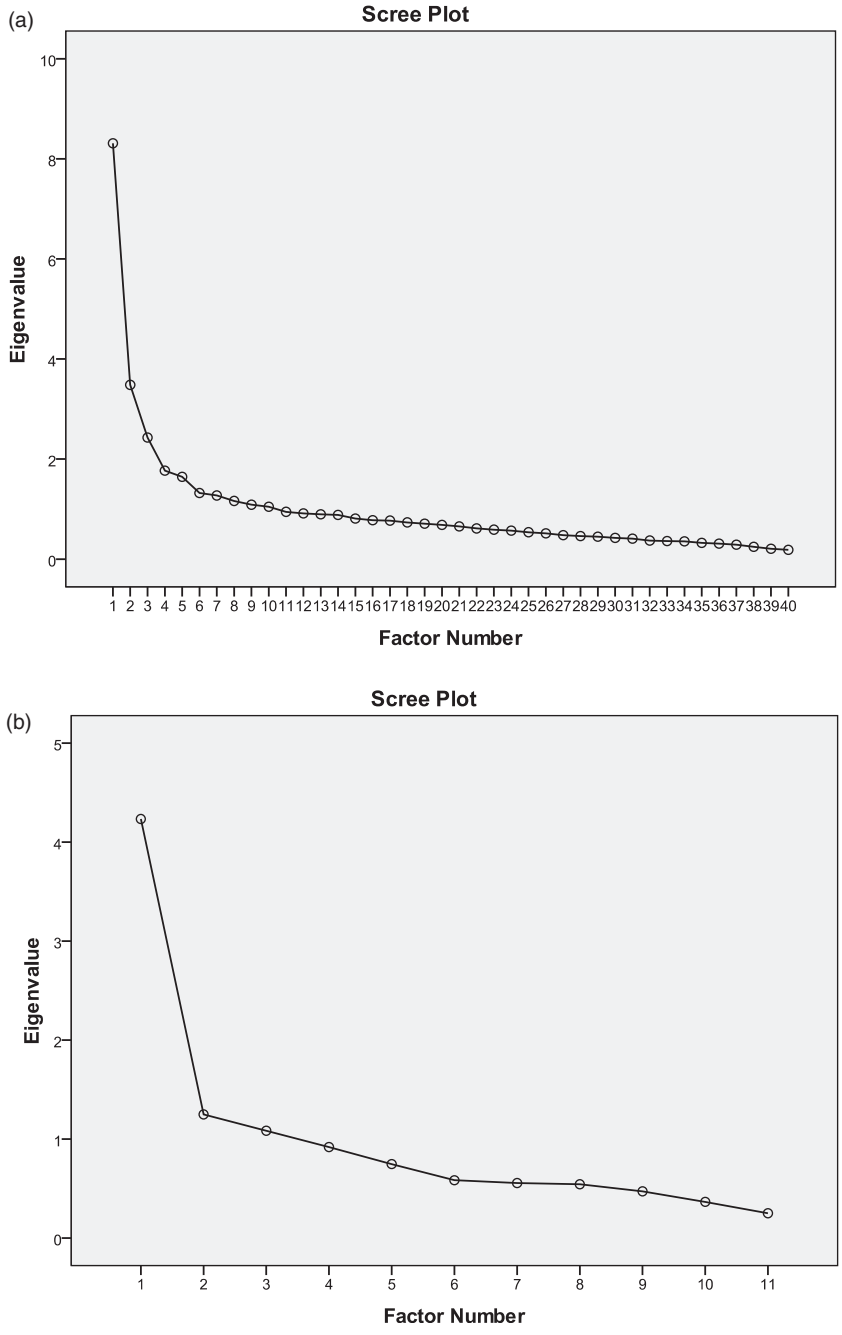


Figure 1. Scree plot of EFA of (a) the original 40 items and (b) the final 11 items.

Table 1. Model fit indices for 3-Factor Patience Questionnaire compared with alternate models.

Description	Chi-sq.	df	AIC	RMSEA	CFI
Final model:					
3-Factor Model (Correlated)	107.30	41	158.25	0.054	0.96
Alternate models:					
3-Factor Model (Orthogonal)	227.49	44	226.63	0.094	0.90
2-Factor Model (Correlated)	186.84	43	246.37	0.100	0.92
2-Factor Model (Orthogonal)	294.07	44	309.42	0.120	0.86
Single-Factor Model	279.19	44	348.99	0.130	0.87

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

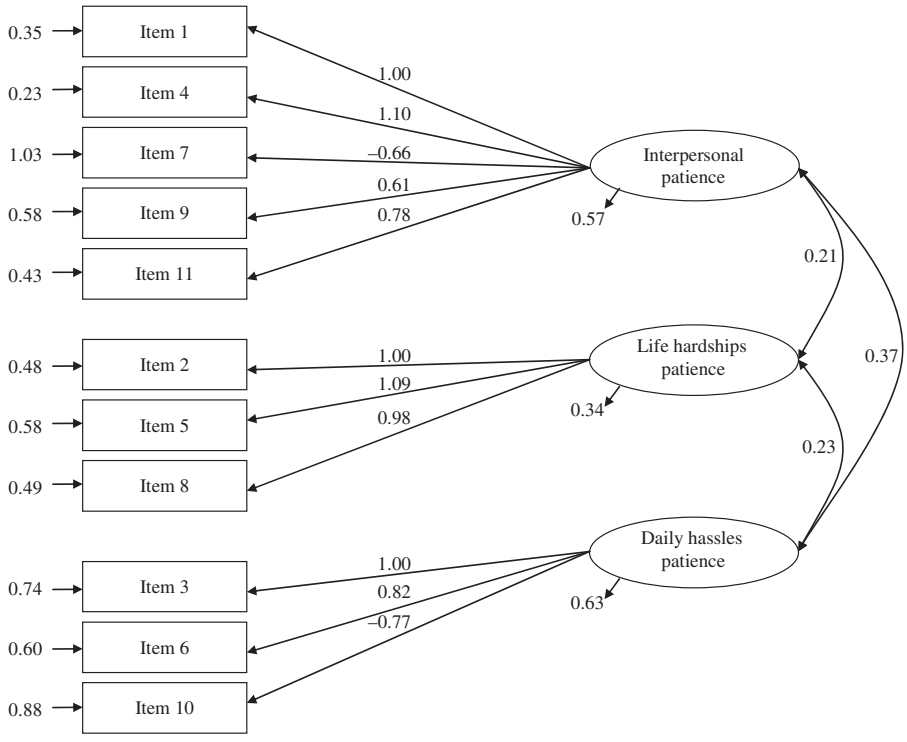


Figure 2. Parameter estimates from confirmatory factor analysis of final measurement model of the 3-FPS (first item loading on each factor set constant to 1).

Table 2. Internal reliability, means, and SDs of the 3-FPS.

	Cronbach's alpha		Test–retest reliability		Mean		SD	
Patience factor	Study 1	Study 2	Study 1	Study 2	Study 1	Study 2	Study 1	Study 2
Interpersonal patience	0.80	0.76	0.74	0.73	3.53	3.56	0.73	0.71
Life hardships patience	0.70	0.76	0.49	0.55	3.19	3.21	0.76	0.79
Daily hassles patience	0.66	0.58	0.75	0.69	3.24	3.32	0.86	0.79

Note: Test–retest reliability is over two weeks for Study 1 and eight weeks for Study 2.

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

Big Five, all five factors are significant predictors, but Agreeableness is the strongest predictor ($\beta = 0.48$) and Neuroticism has a smaller effect ($\beta = -0.17$). Only Neuroticism ($\beta = 0.33$) and Conscientiousness ($\beta = 0.33$) are significant predictors of *life hardship patience*. All the factors, except Openness, significantly predict *daily hassles patience*.

Other antecedent traits. Avoidant attachment is negatively correlated with *interpersonal patience* ($r = -0.26$, $p < 0.01$) and *daily hassles patience* ($r = -0.16$, $p < 0.01$). Anxious attachment is negatively correlated with all three patience factors ($r = -0.22$ to -0.27, $p < 0.01$). Mindfulness is positively correlated with *interpersonal* ($r = 0.27$), *daily hassles* ($r = 0.34$) and *life hardship patience* ($r = 0.11$). When mindfulness is added to a regression equation after controlling for the Big Five, it remains a predictor of only *daily hassles patience*, $\beta = 0.22$, $p < 0.001$; $\Delta R^2 = 0.04$, $F(1,374) = 17.56$, $p < 0.001$. This supports the conceptualization of daily hassles patience as enabling the person to endure frustrations in the present moment.

Differential prediction of well-being outcomes by patience factors

To assess discriminant validity of the three factors of the 3-FPS, the well-being outcomes (life satisfaction, hope, self-esteem, depression, and loneliness) were regressed on the three factors in a series of five regressions to test *a priori* expectations of differential prediction (Table 4). Looking first at loneliness, *interpersonal patience* was hypothesized as its primary predictor since patience with others is integral to maintaining relationships; *interpersonal patience* was the only significant predictor of loneliness ($\beta = -0.15$,

Table 3. Regression of patience on the Big Five.

Personality variable	Total patience score *Beta*	Patience factor *Beta*		
		Regression 1: interpersonal patience	Regression 2: daily hassles patience	Regression 3: daily hassles patience
Extraversion	−0.13**	−0.11**	−0.08	−0.11*
Neuroticism	−0.29**	−0.17**	−0.29**	−0.22**
Agreeableness	0.33**	0.48**	0.07	0.18**
Conscientiousness	0.14**	0.09*	0.13*	0.12*
Openness	0.16**	0.19**	0.09	0.07
R	0.62	0.66	0.40	0.40

Note: *$p < 0.05$, **$p < 0.01$.

Table 4. Regressions predicting well-being indicators from three patience factors.

Patience factor	Well-being indicator *Beta*				
	Regression 1: loneliness	Regression 2: hope	Regression 3: life satisfaction	Regression 4: self-esteem	Regression 5: depression
Interpersonal patience	−0.15*	0.25**	0.13*	0.18**	0.00
Life hardships patience	−0.03	0.22**	0.08	0.13*	−0.04
Daily hassles patience	−0.08	−0.06	0.13*	0.07	−0.19**
R	0.22	0.37	0.27	0.31	0.20

Note: *$p < 0.05$, **$p < 0.01$.

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

$p = 0.01$). Second, we expected *life hardship patience* to influence hope because of its role in creating a positive future outlook. *Life hardship patience* predicted hope ($\beta = -0.22$, $p < 0.001$), as did *interpersonal patience* ($\beta = 0.25$, $p < 0.001$). Next, only *interpersonal* ($\beta = 0.13$, $p = 0.03$) and *daily hassles patience* ($\beta = 0.13$, $p = 0.02$) were significant predictors of life satisfaction, which asks about satisfaction 'so far' in life. *Interpersonal patience* was the strongest predictor of self-esteem ($\beta = 0.18$, $p = 0.002$). Finally, *daily hassles patience* was the primary predictor of depression ($\beta = -0.19$, $p = 0.002$).

Assertiveness and patience

In the prior discussion of the possible negative outcomes of patience, assertiveness was presented as a trait that may be lacking in individuals with 'too much' patience. This was not supported. The zero-order correlations between all three patience factors and assertiveness were not significantly different from zero at the $p < 0.05$ level ($r = -0.06$, 0.04, and -0.02), which indicates independence. Moreover, a scatter plot of the data did not show any curvilinear effects.

Discussion

Study 1 fulfilled its expressed research goals. A new 3-factor survey was created to measure *interpersonal*, *life hardship*, and *daily hassles patience*. Considerable support of construct validity was garnered under John and Soto's (2007) five forms of evidence. Looking at content validity, the 3-FPS is face valid, and its items are representative of trait patience as expressed across different domains. Structural validity is supported by good indicators of fit from the confirmatory factor analyses. The 3-FPS demonstrates both internal reliability and test–retest reliability. Evidence of external validity – both convergent and discriminant – is copious; the 3-FPS relates to other personality traits and well-being outcomes in theoretically expected ways.

Study 2: Patience, goal pursuit, and well-being

The main purpose of Study 2 is to examine patience and well-being in the context of goal pursuit and achievement. Traditionally, researchers examining goal striving have tied certain goal characteristics and content to both hedonic and eudaimonic well-being (McGregor & Little, 1998). For example, goal achievement increases positive affect and hedonic happiness

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

(Ryan & Deci, 2001). Study 2 elicits ten goals from participants and then tracks goal achievement and patience on those goals across time. It is hypothesized that (a) patient people exert more effort in pursuit of their goals and (b) pursue them with more patience than less patient people. Previous research has shown that goal progress across the course of the quarter predicts well-being measured at the end of the quarter and that satisfaction with goal progress directly affects well-being (Sheldon & Elliot, 1999). Similarly, we theorize that patience enacted on goals will increase goal progress/satisfaction, which will then increase well-being.

Study 2 also serves to provide additional evidence of construct validity for patience and the 3-FPS. Reliability and validity indices will replicate Study 1 results, and multimethod assessment with informant reports will check for social-desirability effects.

Method

Participants

Participants were 259 (179 females, 58 males, 11 unknown) UC Davis undergraduates recruited through the department subject pool. They received course credit as remuneration. Data on ethnicity were not collected in order to shorten the length of the surveys. However, participants are assumed to match the general UCD psychology subject pool, which is generally 40–45% Asian American, 40–45% Caucasian, 8–12% Hispanic, and 2–12% Other Ethnicity.

Design

Data were collected five times across the 10-week academic quarter (at weeks 2, 4, 6, 8, and 10). At Times 1 and 5, participants completed a personal projects assessment, as well as measures of patience, personality traits, and well-being variables. At Times 2, 3, and 4, participants completed only the personal projects assessment and well-being measures.

Measures

Patience self-report measure administered at Times 1 and 5

3-Factor Patience Scale. See description under Study 1 Measures.

Personal projects measures

Self-reported projects. Little's (1983) personal projects assessment method was utilized to measure participants' goals. Personal projects are a desirable unit to analyze goals, as they are inherently temporal and interactive (Palys & Little, 1983). At Time 1, participants were given a definition of a personal project and asked to list 10 projects they were working on over the course of the academic quarter (e.g. 'lose ten pounds' or 'complete my thesis').

Project ratings at Times 1–5. Participants were asked to rate their projects on a number of dimensions. Using the previous two weeks as the criterion, they were asked at each time point to rate how much patience each project required and their patience in the pursuit of each project ('This project requires me to be patient as I strive for it' and 'I am patient in my pursuit of this project' 0 = *strongly disagree*; 10 = *strongly agree*). They were also asked to rate each project on variables hypothesized to relate to patience, well-being, and goal achievement. These variables included commitment, control, effort, difficulty, obstacles, time adequacy, progress, expected outcome, satisfaction, competence, expected time frame, meaningfulness, and sanctification. The collection of personal projects over time provides a unique platform to view the effects of patience on multiple levels of analysis.

Well-being measures administered at all time points

CES-D. Participants took the CES-D (see description under Study 1 Measures) to report depressive symptoms for the *past two weeks*.

Satisfaction with Life Scale. See description under Study 1 Measures.

Positive and Negative Affect Schedule. The PANAS (Watson, Clark, & Tellegen, 1988) was used to assess the extent to which subjects felt positive (i.e. interested, enthusiastic, determined, excited, inspired, alert, active, strong, proud, attentive) and negative (i.e. hostile, irritable, guilty, ashamed, nervous, jittery, distressed, upset, afraid, scared) emotions over the *past two weeks* on a 1 (*very slightly or not at all*) to 5 (*extremely*) scale. The PANAS was revised to include more low-arousal positive affect terms (e.g. calm, serene).

Personality measures administered at Times 1 and 5

BFI and self-regulation. See descriptions under Study 1 Measures.

Procedure

Participants were asked to fill out the Time 1 project assessment measure online during the second week of the academic quarter. Every two weeks (i.e. during weeks 4, 6, and 8 of the quarter), they rated their 10 original projects on the rating dimensions and completed well-being questionnaires. During week 10 of the quarter, participants completed the final survey

packet, comprising project ratings, well-being measures, and personality questionnaires.

Informant reports. At Time 1, participants were asked to provide the names and email addresses of three close-others, including, but not limited to, 'a best friend, significant other, person who sees you daily, and/or family member'. Informants were contacted via e-mail and asked to fill out a survey about the participant. The informant survey contained the same personality and well-being measures taken by the participant with the questions reformatted to reference the participant.

Of the 857 informants asked to complete the survey, 533 completed a report about the participant. Of participants, 94% had at least one informant report and 75% had two or more informant reports. The informants were mostly friends (46%), but there were also a substantial number of parents (23%), siblings (16%), and significant others (10%).

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

Results

Construct validation for the 3-FPS

To provide evidence of generalizability, scale statistics for the 3-FPS were calculated with Study 2 data to replicate findings from Study 1. As can be seen in Table 2, current data yielded analogous descriptive statistics, internal reliability, and test–retest reliability coefficients. *Life hardship patience* had a lower test–retest correlation than the other factors in both studies. Rather than instability in the measurement instrument, we suspected that *life hardship patience* judgments are more susceptible to state influences than other types of patience in our college student samples. Additionally, Cronbach's alpha was slightly lower than typical cut-off norms for *daily hassles patience* ($\alpha = 0.58$), but this low alpha was not too alarming considering that the factor contains just three items. Moreover, the factor structure developed in Study 1 was replicated with the new data, and the fit statistics for Study 2 show good fit: RMSEA = 0.046, CFI = 0.98, and $X^2(41,\ N = 260) = 62.67,\ p < 0.001$. Factor loadings and other parameter estimates closely resembled those produced in the original sample.

Informant reports: Evidence of construct validity and social-desirability check

Informant reports were used to formulate a multi-method assessment matrix for patience. Beyond the benefit of adding non-self-report data to the understanding of a trait (Vazire, 2005), informant reports are a preferred way to check social desirability biases that may appear in self-report measurement of positive traits (Piedmont, McCrae, Riemann, & Angleitner, 2000).

Table 5. Participant self-report and informant rating correlations.

	Consensus informant–informant agreement r (N = 192)	Mean informant–participant agreement r (N =243)
3-FPS		
Daily hassles	0.24**	0.25**
Life hardship	0.16*	0.21**
Interpersonal	0.22**	0.35**
Other traits		
Self-regulation	0.39**	0.36**
Well-being variables		
Life satisfaction	0.28**	0.43**
Depression	0.24**	0.28**
Affect		
Positive – low arousal	0.21**	0.27**
Positive – high arousal	0.29**	0.31**
Negative	0.24**	0.32**

Note: *$p < 0.05$, **$p < 0.01$.

Factor structure of informant ratings of patience. The factor structure of the 3-FPS was replicated. Factor loadings for informant reports were congruent to previous samples, and the new fit statistics mirror previous indices, denoting good/fair fit: RMSEA = 0.061, CFI = 0.98, and $X^2(41,\ N = 533) = 121.21,\ p < 0.001$.

Consensus. Consensus among informants, or informant-informant agreement, was calculated from the first two informant ratings collected of participants who had more than one informant ($N = 192$). As can be seen in Table 5, consensus correlations were 0.24, 0.16, and 0.22 for *daily hassles*, *life hardship*, and *interpersonal patience*, respectively. Informants varied in their level of agreement depending on the personality trait of interest. Consensus correlations on patience were similar to those for low arousal positive affect ($r = 0.21$) and depression ($r = 0.24$).

Informant–participant agreement correlations. Agreement between participants' self-reports and informant ratings was examined. The informant ratings for each participant were averaged to create a mean informant rating score per participant for each variable. These mean scores were correlated with the participants' self-reported scores ($N = 243$), such that each participant counted as a case in the agreement calculations. Informant–participant agreement correlations for patience and related variables can be viewed in Table 5. Agreement was lowest for *life hardship patience* and highest for *interpersonal patience*, which

Table 6. Informant and participant correlations between patience factors and life satisfaction/depression.

	Informants[a]		Participants[b]	
	Life satisfaction	Depression	Life satisfaction	Depression
Daily hassles patience	0.34**	−0.25**	0.17**	−0.13*
Life hardship patience	0.35**	−0.27**	0.28**	−0.27**
Interpersonal patience	0.40**	−0.34**	0.26**	−0.17**

Notes: [a]Individual informant ratings, $N = 533$.
[b]$N = 259$.
*$p < 0.05$, **$p < 0.01$.

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

may reflect the fact that information relevant to *interpersonal patience* is highly available to others.

Social desirability. Participants generally rated themselves as lower in patience ($M = 3.40$, standard deviation (SD) = 0.58) than they were rated by their informants ($M = 3.68$, SD = 0.64), $F(1, 790) = 36.14$, $p < 0.001$. This self-diminishment bias is most likely due to the informants' limited accessibility to the participants' internal negative emotionality (the expression of which is often regulated), rather than an actual depression of participant self-ratings from a higher 'true score'.

Correlation between patience and well-being in informants. Informant ratings of participant patience were correlated with informant ratings of participant life satisfaction and depression (Table 6), and these correlations were of equal or greater magnitude than those observed in participant self-reports.

Patient people exert more patience and effort in pursuit of their goals

The first step in looking at patience and goal pursuit was to empirically demonstrate that people high in trait patience pursue goals with more patience and effort than people low in patience. This provides evidence of substantive validity and shows that patient people are not passively disengaged from goals. We examined correlations between trait patience, goal patience, and goal effort. Moderate correlations were found between patience enacted on goals and *interpersonal patience* ($r = 0.31$, $p < 0.01$) as well as *life hardship patience* ($r = 0.24$, $p < 0.01$). There was only a small correlation between *daily hassles patience* and patience enacted on goals ($r = 0.14$, $p < 0.05$). This reflects the goal directed aspects of *interpersonal* and *life hardship patience* versus the more reactive aspects of *daily hassles patience*. A similar pattern of results was found for effort correlated with *interpersonal* ($r = 0.35$, $p < 0.01$), *life hardship* ($r = 0.17$, $p < 0.01$), and *daily hassles* ($r = 0.13$, $p < 0.05$) *patience*.

Likewise, we examined if trait patience predicts the amount of effort enacted on a goal after controlling for basic personality. In Step 1 of a hierarchical linear regression predicting effort enacted on individual goals, the Big Five accounted for 20% of the variance in goal effort. At Step 2, trait patience (measured at Time 1) was entered into the equation. *Interpersonal patience* accounted for significant variance in predicting goal effort after controlling for the Big Five, $\beta = 0.27$, $\Delta R^2 = 0.05$, $F(1, 255) = 14.734$, $p < 0.001$. Neither *life hardship* nor *daily hassles patience* accounted for significant variance in predicting effort after controlling for the Big Five.

Patient people report higher goal achievement satisfaction

Trait patience also predicts goal progress and goal achievement satisfaction. We found moderate zero-order correlations between *interpersonal patience* and goal progress ($r = 0.30$, $p < 0.01$) as well as goal achievement satisfaction ($r = 0.34$, $p < 0.01$; averaged across goals and time). Correlations with goal progress and satisfaction were also found for *life hardship patience* ($r = 0.23$, 0.28, $p < 0.01$). Small correlations with goal progress and satisfaction were found for *daily hassles patience* ($r = 0.13$, 0.15, $p < 0.05$).

We also examined if trait patience predicts goal progress and goal achievement satisfaction after controlling for basic personality. In Step 1 of a hierarchical linear regression predicting goal progress, the Big Five accounted for 17% of the variance in progress. At Step 2, trait patience (measured at Time 1) was entered into the equation. *Interpersonal patience* accounted for significant variance in predicting goal progress after controlling for the Big Five, $\beta = 0.23$, $\Delta R^2 = 0.03$, $F(1, 255) = 9.554$, $p = 0.002$. Neither *life hardship* nor *daily hassles patience* accounted for significant variance in predicting progress after controlling for the Big Five. Similarly, in Step 1 of a hierarchical linear regression predicting goal achievement satisfaction, the Big Five accounted for 20% of the variance in satisfaction. At

Step 2, trait patience was entered into the equation. *Interpersonal patience* accounted for significant variance in predicting goal satisfaction after controlling for the Big Five, $\beta = 0.22$, $\Delta R^2 = 0.03$, $F(1, 255) = 9.691$, $p = 0.002$. Neither *life hardship* nor *daily hassles patience* accounted for significant variance in predicting satisfaction after controlling for the Big Five.

Goal satisfaction mediates the relation between patience and well-being over time

To test the hypothesis that increased goal achievement satisfaction mediates the effect of trait patience at Time 1 on life satisfaction measured at Time 5, nonparametric bootstrapping analyses to compare indirect effects (Preacher & Hayes, 2008) were performed with a 99% bias corrected confidence interval (CI). In bootstrapping, mediation is significant if the CIs do not include zero. Results based on 1000 bootstrapped samples indicated that there was a significant indirect effect for goal satisfaction across time as a mediator of the relation between patience at Time 1 and life satisfaction at Time 5 ($b = 0.30$, lower 99% CI = 0.1154, upper 99% CI = 0.5057). In the mediation model, patience predicted increased goal satisfaction ($b = 0.80$, standard error (SE) = 0.13, $p < 0.001$) and goal satisfaction predicted increased life satisfaction ($b = 0.38$, SE = 0.07, $p < 0.001$).The initial direct effect from patience to life satisfaction ($b = 0.50$, SE = 0.15, $p < 0.001$) was reduced to a non-significant direct after including goal achievement satisfaction in the regression equation ($b = 0.20$, SE = 0.15, $p = 0.192$). Thus, we can say that goal satisfaction nearly fully mediates the relation between patience at Time 1 and life satisfaction at Time 5.

Moderating effects of goal difficulty, obstacles, and patience required

Patience was hypothesized to have a larger effect on goal satisfaction when goals were more difficult and obstacle-ridden, because patience is most essential when people face frustrations. A hierarchical multiple regression was run to test moderation of the relation between patience enacted and goal satisfaction by goal difficulty. In the first step, standardized values of goal satisfaction (ratings entered as specific goals averaged across time alone) were regressed on patience enacted and goal difficulty. Both variables significantly predicted goal satisfaction (Patience Enacted: $\beta = 0.51$, $p < 0.001$; Difficulty: $\beta = -0.40$, $p < 0.001$) and accounted for 41% of the variance in goal satisfaction scores ($R = 0.64$, $p < 0.001$). In the second step of the regression, the interaction term of Patience enacted × Difficulty was significant, $\beta = 0.10$, $\Delta R^2 = 0.01$, $F(1, 2{,}359) = 40.90$, $p < 0.001$. The effect of patience on goal satisfaction is larger for difficult goals. When goals are one SD above the mean on difficulty, moving from low to high patience predicts a 1.18 SD increase in goal satisfaction, but when goals are one SD below the mean on difficulty, moving from low to high patience predicts only a 0.85 SD increase in satisfaction. A similar pattern of results is seen for Patience enacted × Obstacles, $\beta = 0.09$, $\Delta R^2 = 0.01$, $F(1, 2{,}359) = 29.01$, $p < 0.001$, and Patience enacted × Patience required, $\beta = 0.18$, $\Delta R^2 = 0.03$, $F(1, 2{,}359) = 118.65$, $p < 0.001$.

Discussion

Participant and informant data afforded further evidence of construct validity of the 3-FPS. Informant reports supported scale generalizability by demonstrating internal reliability and stable factor structure of the 3-FPS across multiple assessment methods. Informant–participant agreement fell within the range of agreement scores observed for other traits, and the relation between patience and well-being was replicated. Moreover, patient people, as measured by 3-FPS, effortfully pursued their personal projects and enacted more patience as they worked toward their goals. Patience facilitates goal pursuit and satisfaction to improve well-being, and patience is especially crucial for well-being when people are facing difficulties and obstacles.

Study 3: Patience intervention increases well-being

The function of Study 3 is to examine if a patience training intervention can increase trait patience and well-being. The formulation of patience as a character strength connotes that it is a somewhat malleable trait that can be increased by specific practices or cognitive strategies, and increases in the virtue should enhance well-being. The experimental nature of Study 3 allows for testing causality, similar to research designs that have been utilized to examine the well-being effects of other virtues (e.g. gratitude journaling in Emmons & McCullough, 2003)

In this study, a patience training intervention was developed to increase trait patience. Presented as a stress reduction study, activities included group discussions, individual activities, and guided meditation. This study aimed to test two hypotheses: (a) the patience training program increases patience, and (b) the patience training program increases well-being.

Method

Participants

Participants were 71 (61 females, 10 males) UC Davis undergraduates recruited through the department

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

subject pool. They received course credit for remuneration. Participants were primarily Caucasian (40%) and Asian American (34%), though other ethnic minority groups were also represented (11% Hispanic, 7% Middle Eastern, 3% African American, 4% other). Participants were 18–27 years old, with an average age of 20.6 years (SD = 1.8).

Measures

3-Factor Patience Scale. See description under Study 1 Measures.

Satisfaction with Life Scale. See description under Study 1 Measures.

CES-D. Participants took the CES-D (see description under Study 1 Measures) to report depressive symptoms for the *past week*.

PANAS. The PANAS (Watson, Clark, & Tellegen, 1988) was used to assess the extent to which subjects felt positive (e.g. interested, enthusiastic, determined) and negative (e.g. hostile, irritable, guilty) emotions *over the past week* on a 1 (*very slightly or not at all*) to 5 (*extremely*) scale.

Subjective Happiness Scale. Lyubomirsky and Lepper's (1999) four items, rated on a 7-point Likert scale, were used to measure participant happiness. For example, one items reads, 'In general, I consider myself . . . *not a very happy person* (1) to *a very happy person* (7)'.

Gratitude Questionnaire. The GQ-6 (McCullough, Emmons, & Tsang, 2002) is a 6-item measure of gratitude with items such as 'If I had to list everything that I feel grateful for, it would be a very long list' rated on a 1 (*strongly disagree*) to 7 (*strongly agree*) scale.

Emotion regulation. Gross and John's (2003) 10-item Emotion Regulation Questionnaire assesses individual differences in utilizing the two emotion regulation strategies of cognitive reappraisal and expressive suppression (1 = *strongly disagree*; 7 = *strongly agree*).

MAAS. See descriptions under Study 1 Measures.

Procedure

Upon agreement to participate in the study, which was presented as a stress and personality study, participants attended a 45-minute introductory session and completed the study measures. They were then randomly assigned to the experimental or control condition.

Participants in the *experimental condition* (i.e. the patience training intervention) signed up for four training sessions to be held across the following two weeks. After the last training session, participants were instructed to complete an online survey, retaking the well-being and personality measures from Time 1. Participants in the *control group* took the Time 1 survey at the initial training session and were sent a link for the Time 2 survey when it was sent to the experimental group. One month following the second survey, all participants were e-mailed a link for a third survey, also containing the original measures.

Experimental group patience training sessions. The patience training was administered by the experimenter to groups of three to six participants over four 30-minute sessions. The training consisted of teaching and activity components, and it was formulated by drawing upon interventions and methods utilized in prior studies, specifically deriving insight from the meditation, Type A, self-control, CBT, and character strengths literatures.

Didactic component. The first session was focused on increasing participants' awareness of their positive and negative emotions, as well as the precipitating causes and triggers of those emotions. Meditative practice was also introduced and discussed in the first session. The second session focused on emotion regulatory strategies, both preventive and reactionary. The third and fourth sessions were focused on coping with interpersonal stress and fostering a sense of empathy and compassion. Participants reflected upon past situations in which they were frustrated with another person and answered questions meant to encourage perspective-taking. Participants also completed a reframing/reappraisal exercise. They were asked to consider a variety of frustrating situations and write their typical negative thoughts before writing a reappraisal of the situation. For instance, participants were given the following scenario,

> *You are catching a new release at the movies. The person behind you keeps kicking your chair and chewing her popcorn really loudly. As the movie progresses, she proceeds to answer her phone twice*,

and asked to describe their typical negative response (e.g. '*People are so annoying! Can't they understand that I'm trying to watch a movie?*'), as well as the reframe (e.g. '*I've probably accidently kicked the chair in front of me before, too, so maybe it was an accident*').

Guided meditation. Each session contained a 10–15-minute guided meditation. Meditations began with instructions for participants to attend to their breathing and posture in a non-judgmental manner. During the first two training sessions, the meditation was focused on replacing negative emotions with positive emotions. The third and fourth sessions included a loving-kindness meditation. Loving-kindness

meditation involves directing warm and positive emotions toward other people, beginning with the self and extending gradually outward (i.e. to a benefactor, good friend, neutral person, enemy, all beings of the world; Carson et al., 2005; Fredrickson, Cohn, Coffery, Pek, & Finkel, 2008; Kristeller & Johnson, 2005). The loving-kindness meditation was included due to the highly interpersonal nature of patience.

Results

One-way analyses of variance indicate there were no differences between the experimental and control groups at Time 1 on all study variables (at $p < 0.05$).

Effect of patience training on Time 2 variables controlling for Time 1 scores

Patience training increases trait patience. *Interpersonal patience* at Time 2 was regressed on *interpersonal patience* at Time 1 and accounted for 54% of the variance in Time 2 patience scores ($\beta = 0.733$, $p < 0.001$). At Step 2, treatment group (dummy coded: 0 = control, 1 = experimental) accounted for significant variance in Time 2 scores, $\beta = 0.21$, $\Delta R^2 = 0.05$, $F(1, 54) = 5.65$, $p = 0.02$. Participation in the treatment intervention predicted higher *interpersonal patience* scores at Time 2, controlling for Time 1 patience. Treatment did not predict Time 2 *life hardship patience*, $\beta = 0.08$, $\Delta R^2 = 0.01$, $F(1, 54) = 0.54$, $p = 0.47$, nor did it predict *daily hassles patience*, $\beta = 0.16$, $\Delta R^2 = 0.02$, $F(1, 54) = 2.70$, $p = 0.11$.

Patience training predicts changes in positive, but not negative, affect. Time 2 positive affect was regressed on Time 1 positive affect, which accounted for 48% of the variance in Time 2 scores ($\beta = 0.69$, $p < 0.001$). At Step 2, treatment group accounted for significant variance, $\beta = 0.20$, $\Delta R^2 = 0.04$, $F(1, 54) = 4.37$, $p = 0.04$, such that participation in the intervention predicted higher positive affect at Time 2. Treatment group did not predict Time 2 negative affect controlling for Time 1 negative affect, $\beta = -0.02$, $\Delta R^2 < 0.001$, $F(1, 54) = 0.02$, $p = 0.88$.

Patience training decreases depression. Time 2 depression was regressed on Time 1 depression, which accounted for 15% of the variance in Time 2 depression scores ($\beta = 0.39$, $p = 0.004$). At Step 2, treatment group accounted for significant variance in scores, $\beta = -0.26$, $\Delta R^2 = 0.067$, $F(1, 54) = 4.53$, $p = 0.04$, such that patience training significantly decreased depression.

Patience training and emotion regulation. Time 2 reappraisal was regressed on Time 1 reappraisal, which accounted for 44% of the variance in Time 2 scores ($\beta = 0.66$, $p < 0.001$). Treatment group then accounted for marginally significant variance, $\beta = 0.19$, $\Delta R^2 = 0.03$, $F(1, 54) = 3.41$, $p = 0.07$. Patience training may increase participants' use of reappraisal. Treatment had no effect on suppression, $\beta = 0.12$, $\Delta R^2 = 0.012$, $F(1, 54) = 1.03$, $p = 0.31$.

Life satisfaction, happiness, gratitude, and mindfulness. Effects of the intervention at Time 2 did not reach statistical significance for life satisfaction, $\beta = 0.13$, $\Delta R^2 = 0.02$, $F(1, 54) = 1.77$, $p = 0.19$; happiness, $\beta = 0.10$, $\Delta R^2 = 0.01$, $F(1, 54) = 0.93$, $p = 0.34$; gratitude, $\beta = 0.09$, $\Delta R^2 = 0.01$, $F(1, 54) = 0.86$, $p = 0.36$; or mindfulness, $\beta = -0.001$, $\Delta R^2 < 0.001$, $F(1, 54) = 0.001$, $p = 0.94$.

Mediation effects

To test the hypothesis that increased patience mediates the effect of the patience intervention on well-being measured at Time 2, nonparametric bootstrapping analyses to compare indirect effects with covariates (Preacher & Hayes, 2008) were performed. Given the small sample size and exploratory approach, 90% bias corrected CIs were examined for the indirect effects (mediation is significant if the CI does not include zero).

There was partial support that *interpersonal patience* mediates the effect of the patience training intervention on T2 depression. Results based on 1000 bootstrapped samples indicated that there was a significant indirect effect for *interpersonal patience* ($b = -0.0193$, lower 90% CI = -0.0855, upper 90% CI = -0.0001), controlling for T1 depression. Although there seems to be a significant indirect effect, the path coefficients from the intervention to interpersonal patience and from interpersonal patience to depression do not reach statistical significance. Given the small sample size of the mediation analysis ($N = 55$), this may be from a lack of power. Therefore, we cannot conclude that interpersonal patience mediates the effect of the patience training, but neither should we reject the mediating effect. Instead, future studies are needed to clarify the effect with more statistical power.

Patience was also tested as a mediator of the intervention's effect on T2 positive affect. Results based on 1000 bootstrapped samples indicated that there was no significant indirect effect for *interpersonal patience* ($b = -0.0193$, lower 90% CI = -0.0090, upper 90% CI = 0.1235).

Effect of patience training on Time 3 variables controlling for Time 1 scores

Time 3 trait patience. *Interpersonal patience* at Time 3 was regressed on *interpersonal patience* at Time 1, which accounted for 39% of the variance in Time 3

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

scores ($\beta = 0.622$, $p < 0.001$). Treatment group was then added to the equation, $\beta = 0.230$, R^2 change $= 0.049$, $F(1,42) = 3.475$, $p = 0.07$. Even though this effect only reaches marginal significance, the beta coefficient is actually larger than the coefficient for predicting patience at Time 2 ($\beta = 0.21$).

Although there was no effect of treatment on *life hardship patience* at Time 2, there was a marginally significant effect of treatment on *life hardship patience* at Time 3. *Life hardship patience* at Time 1 accounted for 27% of the variance in Time 3 *life hardship patience* scores ($\beta = 0.52$, $p < 0.001$). At Step 2, treatment group accounted for a marginally significant variance not explained by Time 1 scores, $\beta = 0.24$, $\Delta R^2 = 0.06$, $F(1, 42) = 3.49$, $p = 0.07$. The beta is substantially larger than was given for predicting *life hardship patience* at Time 2 ($\beta = 0.08$). This evidence of a delayed effect of the patience training on *life hardship patience* fits *a priori* hypotheses that participants will only become aware of changes in their *life hardship patience* when they have sufficient opportunity to observe themselves reacting to long-term stressors over time.

There was no effect of the intervention on ratings of *daily hassles patience* at Time 3, controlling for Time 1 scores, $\beta = -0.05$, $\Delta R^2 = 0.003$, $F(1, 42) = 0.23$, $p = 0.64$.

Time 3 depression. Time 3 depression was regressed on Time 1 depression, which accounted for 6% of the variance in Time 3 scores ($\beta = 0.25\, p = 0.11$). Treatment group was then added to the regression equation, $\beta = -0.19$, $\Delta R^2 = 0.03$, $F(1, 42) = 1.51$, $p = 0.23$. Although treatment was not a significant predictor of Time 3 depression, the beta coefficient was similar in size to the beta for predicting Time 2 depression ($\beta = -0.26$). Also, Time 1 depression scores failed to reach statistical significance for predicting depression at Time 3, which indicates a power issue.

Discussion

Study 3 provided evidence in partial support of its two hypotheses. First, levels of trait patience were immediately raised by the patience training intervention, and results from a one month follow up point to a sustained increase in patience (even though results do not reach significance because of subject attrition). These results support the conceptualization of patience as a character strength that can be strengthened with specific practices and activities.

The second hypothesis was also supported: the patience training program increased well-being. Controlling for Time 1 scores, participants who partook in the patience training reported lower depression and higher positive affect at Time 2. Time 3 effects were not significant, but the pattern of results suggests they would most likely reach significance with higher power.

As this was the first attempt to directly test if trait patience could be increased through intentional activities, the patience training intervention was constructed to minimize Type II errors by including a broad range of exercises and activities. With initial results supporting the efficacy of the intervention, the separable effects of the particular components (meditation, cognitive reappraisal training, etc.) can now be examined to see which activities (or combination of them) are driving the increase in patience. A limitation resulting from the broad scope of the patience training exercises is that we cannot exclusively attribute the well-being effects of the manipulation to increased patience. The intervention could also be enhancing related psychological attributes, such as adaptive coping, mindfulness, emotion regulation, and compassion. However, the non-significant effects of the intervention on Time 2 and Time 3 mindfulness and gratitude suggest that the intervention is increasing patience – not other virtues!

While some may propose that our control condition does not rule out the possibility that our effects are primarily the result of demand characteristics, such an explanation is not supported by the pattern of results. If demand characteristics were driving our effects, we would expect the patience training to increase scores on all of the well-being outcomes equally. Instead, we observe that the training alleviates depression, but does not significantly change life satisfaction or happiness. Of these three well-being indicators, depression is least susceptible to demand characteristics (with non-intuitive items such as 'I did not feel like eating; my appetite was poor' and 'My sleep was restless'), whereas the life satisfaction and happiness are more susceptible to demand characteristics (with transparent items such as 'I am satisfied with my life' and 'I consider myself a happy person'). Future studies with additional control conditions (e.g. a control condition that teaches organization skills) will further rule out this alternative.

General discussion

Studies 1–3 fulfill their three research objectives: they provide evidence of construct validity for the 3-FPS, longitudinal and experimental data support that patience increases well-being, and longitudinal data support mediation of the patience–well-being relation by goal pursuit and satisfaction.

Hedonic and eudaimonic well-being

Patience consistently relates to indicators of hedonic well-being. Even after controlling for personality,

patience predicts depression, life satisfaction, happiness, and affect in both self- and informant-reports. Goal achievement satisfaction mediates the relation between patience and life satisfaction, supporting the hypothesis that patience increases hedonic well-being through increased goal achievement. Future studies should be designed to collect more objective measures of goal achievement (i.e. grades, relationship outcomes, performance on a laboratory task, etc.) in addition to participant ratings of goal achievement, which are confounded with achievement satisfaction. Objective measures will distinguish if patience increases goal satisfaction because (a) it increases actual goal progress/achievement, or (b) it changes the standards by which people evaluate their goal progress and construct satisfaction judgments.

Patience also relates to eudaimonic well-being. Higher self-esteem in patient people points to higher fulfillment of their competence and autonomy needs, and lower loneliness points to the fulfillment of relatedness needs. In addition, patient people score higher on measures of other virtues that are a part of the eudaimonic notion of 'the good life'.

Patience does not lead to ill-being. In addition to predicting well-being, patience does not predict negative outcomes, nor is there evidence that 'too much' patience is maladaptive. Instead, data support Kierkegaard's assertion that 'patience is not resignation, passivity, or inaction; rather, it is the emergence of freedom within the domain where necessity rules' (Harned, 1997, p. 101). Study 1 results show that patience and assertiveness are orthogonal, and Study 2 shows that patient people pursue goals with more effort than less patient people.

State and trait patience

Patience was originally presented both as a trait and a state. Although the trait formulation of the virtue received the most attention in the three studies, a more state-like form of patience was also examined: patience enacted on specific goals (Study 2). Patience enacted on goals was moderately correlated with trait patience, but there was considerable unshared variance between the two. Future studies could use a multileveling modeling framework (of goals nested within persons) to examine this state-trait connection.

Moderation of the patience–well-being relation

In Study 2, the difficulty of goals moderated the impact of patience enacted on goal satisfaction, such that patience played a more prominent role in predicting satisfaction for goals that were difficult and obstacle-ridden. Studies should further examine the moderating role of situation characteristics. Patience, like secondary control, may be most adaptive when the person is unable to exert an influence on the environment and primary control efforts prove futile (Lam & Zane, 2004). For example, Rothermund and Brandtstadter (2003) found that switching to an accommodative strategy of flexible goal adjustment mitigated the negative impact of performance declines on self-evaluations in older people.

On the flip side, are there certain situations in which patience may lead to negative outcomes? Is patience a maladaptive response when the person does have control over the environment? If such situations are found, the distinction between state and trait patience may play into our understanding of the virtue. Moral philosophers would argue that trait patience is always beneficial and required for effective action, because only when a person is dispositionally patient can he or she clearly recognize when a situation requires an elevated emotional response (low state patience) and when to wait calmly (high state patience; Harned, 1997). From this perspective, high state patience – but not high trait patience – could lead to negative outcomes in particular situations. Such an argument would apply to positive traits and virtues more broadly, and research in this area would deepen our understanding of person–situation interactions.

Acknowledgments

I would like to express gratitude to Dr Robert Emmons for his guidance in this research.

References

Block, J. (1996). Some jangly remarks on Baumeister and Heatherton. *Psychological Inquiry, 7*, 28–32.

Brandtstadter, J., & Renner, G. (1990). Tenacious goal pursuit and flexible goal adjustment: Explication and age-related analysis of assimilative and accommodative strategies of coping. *Psychology and Aging, 5*, 58–67.

Brandtstadter, J., & Rothermund, K. (2002). The life-course dynamics of goal pursuit and goal adjustment: A two-process framework. *Developmental Review, 22*, 117–150.

Brennan, K.A., Clark, C.L., & Shaver, P.R. (1998). Self-report measurement of adult attachment: An integrative overview. In J.A. Simpson & W.S. Rholes (Eds.), *Attachment theory and close relationships* (pp. 46–76). New York, NY: Guilford Press.

Britons spent six months queuing: Britons will spend almost six months of their life queuing. (2009, March 26). The Daily Telegraph. Retrieved from http://www.telegraph.co.uk/news/newstopics/howaboutthat/5052956/Britons-spent-six-months-queuing.html

Brown, K.W., & Ryan, R.M. (2003). The benefits of being present: Mindfulness and its role in psychological well-being. *Journal of Personality and Social Psychology, 84*, 822–848.

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

Buehler, R., Griffin, D., & MacDonald, H. (1997). The role of motivated reasoning in optimistic time predictions. *Personality and Social Psychology Bulletin, 23*, 238–247.

Burke, E. (1769). *Observations on a late state of the nation*. London, UK: J. Dodsley.

Byram, S.J. (1997). Cognitive and motivational factors influencing time prediction. *Journal of Experimental Psychology: Applied, 3*, 216–239.

Byrne, B.M. (1998). *Structural equation modeling with Lisrel, Preslis, and Simplis: Basic concepts, applications, and programming*. Mahwah, NJ: Lawrence Erlbaum Associates.

Campbell, D.T., & Fiske, D.W. (1959). Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological Bulletin, 56*, 81–105.

Carson, J.W., Keefe, F.J., Lynch, T.R., Carson, K.M., Goli, V., Fras, A.M., & Thorp, S.R. (2005). Loving-kindness meditation for chronic low back pain: Results from a pilot trial. *Journal of Holistic Nursing, 23*, 287–304.

Clore, G.L., & Robinson, M.D. (2000). What is emotion regulation? In search of a phenomenon. *Psychological Inquiry, 11*, 163–166.

Curry, O.S., Price, M.E., & Price, J.G. (2008). Patience is a virtue: Cooperative people have lower discount rates. *Personality and Individual Differences, 44*, 780–785.

DeLongis, A., Coyne, J.C., Dakof, G., Folkman, S., & Lazarus, R.S. (1982). Relationship of daily hassles, uplifts, and major life events to health status. *Health Psychology, 1*, 119–136.

Diener, E., Emmons, R.A., Larsen, R.E., & Griffin, S. (1985). The satisfaction with life scale. *Journal of Personality Assessment, 49*, 71–75.

Diener, E., Sapyta, J.J., & Suh, E. (1998). Subjective well-being is essential to well-being. *Psychological Inquiry, 9*, 33–37.

Echteld, M.A., Elderen, T.M.T., & van der Kamp, L.J.T. (2001). How goal disturbance, coping and chest pain relate to quality of life: A study among patients waiting for PTCA. *Quality of Life Research, 10*, 487–501.

Emmons, R.A., & McCullough, M.E. (2003). Counting blessings versus burdens: An experimental investigation of gratitude and subjective well-being in daily life. *Journal of Personality and Social Psychology, 84*, 377–389.

Fredrickson, B.L., Cohn, M.A., Coffery, K.A., Pek, J., & Finkel, S.M. (2008). Open hearts build lives: Positive emotions, induced through loving-kindness meditation, build consequential personal resources. *Journal of Personality and Social Psychology, 95*, 1045–1062.

Glicksohn, J. (2001). Temporal cognition and the phenomenology of time: A multiplicative function for apparent duration. *Consciousness and Cognition, 10*(1), 1–25.

Gross, J.J., & John, O.P. (2003). Individual differences in two emotion regulation processes: Implications for affect, relationships, and well-being. *Journal of Personality and Social Psychology, 85*, 348–362.

Groth, M., & Gilliland, S.W. (2006). Having to wait for service: Customer reactions to delays in service delivery. *Applied Psychology: An International Review, 55*, 107–129.

Harned, D.B. (1997). *Patience: How we wait upon the world*. Cambridge, MA: Cowley Publications.

Hughes, M.E., Waite, L.J., Hawkley, L.C., & Cacioppo, J.T. (2004). A short scale for measuring loneliness in large surveys. *Research on Aging, 26*, 655–672.

John, O.P., Donahue, E.M., & Kentle, R.L. (1991). *The big five inventory—versions 4a and 54*. Berkeley, CA: UC Berkeley, Institute of Personality and Social Research.

John, O.P., & Soto, C.J. (2007). The importance of being valid: Reliability and the process of construct validation. In R.W. Robins, R.C. Fraley, & R.F. Krueger (Eds.), *Handbook of research methods in personality psychology* (pp. 461–494). London: Guilford Press.

Kristeller, J.L., & Johnson, T. (2005). Cultivating loving kindness: A two-stage model of the effects of meditation on empathy, compassion and altruism. *Zygon, 40*, 391–407.

Lam, A.G., & Zane, N.W.S. (2004). Ethnic differences in coping with interpersonal stressors: A test of self-construals as cultural mediators. *Journal of Cross-Cultural Psychology, 35*, 446–459.

Little, B.R. (1983). Personal projects: A rationale and method for investigation. *Environment and Behavior, 15*, 273–309.

Lyubomirsky, S., & Lepper, H.S. (1999). A measure of subjective happiness: Preliminary reliability and construct validation. *Social Indicators Research, 46*, 137–155.

McCullough, M.E., Emmons, R.A., & Tsang, J. (2002). The grateful disposition: A conceptual and empirical topography. *Journal of Personality and Social Psychology*, *82*, 112–127.

McGregor, I., & Little, B.R. (1998). Personal projects, happiness, and meaning: On doing well and being yourself. *Journal of Personality and Social Psychology, 74*, 494–512.

Miller, T.Q., Smith, T.W., Turner, C.W., Guijarro, M.L., & Hallet, A.J. (1996). Meta-anlaytic review of research on hostility and physical health. *Psychological Bulletin, 119*, 322–348.

Palys, T.S., & Little, B.R. (1983). Perceived life satisfaction and the organization of personal project systems. *Journal of Personality and Social Psychology, 44*, 1221–1230.

Peterson, C., Park, N., & Seligman, M.E.P. (2005). Assessment of character strengths. In G.P. Koocher, J.C. Norcross, & S.S. Hill III (Eds.), *Psychologists' desk reference* (2nd ed., pp. 93–98). New York, NY: Oxford University Press.

Peterson, C., & Seligman, M.E.P. (2004). *Character strengths and virtues: A handbook and classification*. Washington, DC: American Psychological Association.

Piedmont, R.L., McCrae, R.R., Riemann, R., & Angleitner, A. (2000). On the invalidity of validity scales: Evidence from self-reports and observer ratings in volunteer samples. *Journal of Personality and Social Psychology, 78*, 582–593.

Preacher, K.J., & Hayes, A.F. (2008). Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behavior Research Methods*, *40*, 879–891.

Radloff, L.S. (1977). The CES-D Scale: A self-report depression scale for research in the general population. *Applied Psychological Measurement, 1*, 385–401.

Rathus, S.A. (1973). A 30-item schedule for assessing assertive behavior. *Behavior Therapy, 4*, 398–406.

Robins, R.W., John, O.P., Caspi, A., Moffitt, T.E., & Stouthamer-Loeber, M. (1996). Resilient, overcontrolled, and undercontrolled boys: Three replicable personality types. *Journal of Personality and Social Psychology, 70*, 157–171.

Rorty, A. (1980). *Essays on Aristotle's ethics*. Berkeley: University of California Press.

Rose, G.M., Meuter, M.L., & Curran, J.M. (2005). On-line waiting: The role of download time and other important predictors on attitude toward E-retailers. *Psychology & Marketing. Special Issue: Experiments in E-commerce, 22*, 127–151.

Rosenberg, M. (1965). *Society and the adolescent self-image*. Princeton, NJ: Princeton University Press.

Rothermund, K., & Brandtstadter, J. (2003). Coping with deficits and losses in later life: From compensatory action to accommodation. *Psychology and Aging, 18*, 896–905.

Roy, M.M., Christenfeld, N.J.S., & McKenzie, C.R.M. (2005). Underestimating the duration of future events: Memory incorrectly used or memory bias? *Psychological Bulletin, 131*, 738–756.

Ryan, R.M., & Deci, E.L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist, 55*, 68–78.

Ryan, R.M., & Deci, E.L. (2001). On happiness and human potentials: A review of research on hedonic and eudaimonic well-being. *Annual Review of Psychology, 52*, 141–166.

Ryff, C.D., & Singer, B. (1998). The contours of positive human health. *Psychological Inquiry, 9*, 1–28.

Schnitker, S.A., & Emmons, R.A. (2007). Patience as a virtue: Religious and psychological perspectives. *Research in the Social Scientific Study of Religion, 18*, 177–207.

Sheldon, K.M., & Elliot, A.J. (1999). Goal striving, need satisfaction and longitudinal well-being: The self-concordance model. *Journal of Personality and Social Psychology, 76*, 482–497.

Sheldon, K.M., Ryan, R.M., Deci, E.L., & Kasser, T. (2004). The independent effects of goal contents and motives on well-being: It's both what you pursue and why you pursue it. *Personality and Social Psychology Bulletin, 30*, 475–486.

Snyder, C.R., Harris, C., Anderson, J.R., Holleran, S.A., Irving, L.M., Sigmon, S.T., & Harney, P. (1991). The will and the ways: Development and validation of an individual differences measure of hope. *Journal of Personality and Social Psychology, 60*, 570–585.

Spence, J.T., Helmreich, R.L., & Pred, R.S. (1987). Impatience versus achievement strivings in the Type A pattern: Differential effects on students' health and academic achievement. *Journal of Applied Psychology, 72*, 522–528.

Vazire, S. (2005). Informant reports: A cheap, fast, and easy method for personality assessment. *Journal of Research in Personality, 40*, 472–481.

Watson, D., Clark, L.A., & Tellegen, A. (1988). Development and validation of brief measures of positive and negative affect: The PANAS scales. *Journal of Personality and Social Psychology, 54*, 1063–1070.

Wolk, S. (1976). Situation constraint as a moderator of the locus of control-adjustment relationship. *Journal of Consulting and Clinical Psychology, 44*, 420–427.

Downloaded by [Sarah Schnitker] at 19:09 26 June 2012

Appendix: *3-Factor Patience Questionnaire*

Instructions: For each of the statements below, please indicate how much the statement is like/unlike you.

1 = Not like me at all
2 = Unlike me
3 = Neutral
4 = Like me
5 = Very much like me

___ 1. My friends would say I'm a very patient friend.
___ 2. I am able to wait-out tough times.
___ 3. Although they're annoying, I don't get too upset when stuck in traffic jams.
___ 4. I am patient with other people.
___ 5. I find it pretty easy to be patient with a difficult life problem or illness.
___ 6. In general waiting in lines does not bother me.
___ 7. I have trouble being patient with my close friends and family.
___ 8. I am patient during life hardships.
___ 9. When someone is having difficulty learning something new, I will be able to help them without getting frustrated or annoyed.
___ 10. I get very annoyed at red lights.
___ 11. I find it easy to be patient with people.

Factor 1 – Interpersonal patience: 1, 4, 7(r), 9, 11
Factor 2 – Life hardship patience: 2, 5, 8
Factor 3 – Daily hassles patience: 3, 6, 10(r)